
SPECIFICATION — HARMONY ENGINE

CROSS-ARCHITECTURE PHYSIOLOGICAL CORRELATION ENGINE FOR MULTI-NODE ALIGNMENT SCORING AND PRIVACY-PRESERVING COMPATIBILITY ANALYSIS (NON-LIMITING)

A. Technical Field

This disclosure relates to physiological data correlation systems and, more particularly, to a cross-architecture correlation engine configured to align multi-node physiological datasets acquired from two or more users with differing physiological architectures (e.g., masculine and feminine body types) for the purpose of generating privacy-preserving compatibility analyses, alignment scores, and longitudinal resonance profiles.

B. Background and Motivation (Non-Limiting)

Existing physiological monitoring systems typically analyze data from a single user in isolation. Where cross-user analysis exists, it is generally limited to comparing identical sensor modalities (e.g., comparing one heart rate to another). No known system provides a principled framework for correlating physiological signals acquired from fundamentally different sensing architectures designed for different anatomical contexts (e.g., penile rigidity signals correlated with intravaginal pressure distribution patterns). The Harmony Engine addresses this gap by providing a unified correlation layer that operates across architecture boundaries while preserving privacy by operating exclusively on abstracted signal representations.

C. Overview (Non-Limiting)

In certain embodiments, the Harmony Engine is a software correlation layer that receives abstracted physiological data from two or more node ecosystems belonging to different users. Each user's data has already been processed through per-node abstraction pipelines (as described in the system specification) to produce indices, zones, trends, and confidence indicators rather than raw sensor signals. The Harmony Engine then performs cross-architecture alignment analysis comprising:

- Multi-dimensional pattern correlation across abstracted signal domains
- Temporal alignment of longitudinal trends between two users
- Generation of composite alignment scores across one or more physiological dimensions
- Privacy-preserving output in the form of indices, symbolic representations, or abstract scores
- Zero-linkage integrity ensuring outputs cannot be reverse-engineered to anatomical data

D. Cross-Architecture Correlation Methods (Non-Limiting)

In various embodiments, the Harmony Engine correlates data across differing physiological architectures using one or more of the following methods:

1. **Dimensional Mapping:** Abstracted metrics from each architecture are mapped into a common feature space. For example, rigidity dynamics from Node 1 and pressure distribution patterns from Node 4 may each be projected into a normalized multi-dimensional space where correlation can be computed.
2. **Temporal Phase Alignment:** Longitudinal trends from each user are temporally aligned to identify co-occurring patterns, convergent trends, or complementary rhythms across differing physiological domains.
3. **Pattern Signature Extraction:** Each user's multi-node data is reduced to a compact pattern signature comprising normalized indices, weighted vectors, or latent representations that capture characteristic physiological behavior without exposing raw values.
4. **Cross-Domain Similarity Scoring:** Similarity or complementarity metrics are computed between pattern signatures using distance measures, cosine similarity, or learned correlation functions.
5. **Contextual Conditioning:** Alignment scores may be conditioned on systemic context (e.g., cardiovascular state, autonomic balance, sleep quality) provided by the upper-body context node (Node 5) to improve interpretation accuracy.

E. Alignment Scoring and Multi-Dimensional Output (Non-Limiting)

In certain embodiments, the Harmony Engine outputs one or more alignment scores representing compatibility or resonance between two users' physiological profiles. These scores may comprise:

- A composite alignment index representing overall multi-dimensional compatibility
- Per-dimension sub-scores (e.g., timing alignment, intensity alignment, rhythm alignment)
- Confidence indicators reflecting data quality, coverage, and temporal depth
- Trend indicators showing whether alignment is improving, stable, or diverging over time
- Zone-based summaries indicating regions of strong or weak alignment

In various embodiments, alignment scores are computed using only abstracted pattern signatures and never from raw sensor data. The scoring system is designed such that no individual user's raw physiological measurements can be inferred or reconstructed from the alignment output.

F. Privacy Architecture and Zero-Linkage Integrity (Non-Limiting)

In various embodiments, the Harmony Engine enforces the following privacy constraints:

1. Abstraction-Only Input: The engine receives only pre-abstracted indices and pattern signatures, never raw sensor streams or waveform data.
2. Non-Invertible Transformation: Pattern signatures are generated through non-invertible transformations such that raw data cannot be reconstructed from the signature.
3. Mutual Consent Gating: Cross-user correlation is only performed when both users have explicitly consented to the specific comparison level.
4. Progressive Disclosure: Alignment results may be revealed in progressive detail levels, with deeper analysis requiring escalating mutual consent.
5. Zero-Linkage Output: All outputs (scores, indices, symbolic representations) are designed so that no anatomical characteristic, raw measurement, or personally identifiable physiological feature can be inferred.
6. Local Processing Option: In certain embodiments, cross-user correlation may be computed locally on a user's device using encrypted profile exchanges, avoiding centralized data aggregation.

G. Longitudinal and Temporal Correlation (Non-Limiting)

In certain embodiments, the Harmony Engine supports longitudinal correlation by maintaining temporal histories of alignment scores and pattern signatures. This enables:

- Tracking how alignment between two users evolves over weeks, months, or longer
- Identifying convergent or divergent physiological trends between paired users
- Detecting transient alignment shifts associated with health changes or external factors
- Generating alignment trajectory projections based on historical patterns

In various embodiments, longitudinal data is stored in encrypted, versioned profile segments. Each segment represents a temporal window and contains only abstracted representations.

H. Integration with Multi-Node Ecosystem (Non-Limiting)

In various embodiments, the Harmony Engine integrates with the following node types and data sources:

- Node 1 (External Longitudinal Wearable): Rigidity dynamics, stability indices, response timing, and physiological recovery patterns
- Node 2 (Intravaginal Longitudinal Wearable): Impedance-derived tissue state, hydration/moisture trends, pelvic activity indices, and pH trends
- Node 3 (External Geometry Scanner): Baseline geometry references, zone profiles, and repeatability indices
- Node 4 (Intraluminal Device): Internal geometry, compliance maps, contact distribution, and expansion indices
- Node 5 (Upper-Body Context Node): Cardiovascular load, autonomic balance, sleep context, and activity patterns

Cross-architecture correlation may use any subset of available node data. The engine is designed to produce meaningful alignment scores even when only partial node coverage is available for one or both users.

I. Exemplary Claim-Like Statements (Non-Limiting)

A provisional application does not require claims. The following statements are illustrative only and do not limit the disclosure.

1. A system comprising: a) a first data abstraction pipeline configured to receive sensor data from one or more physiological sensing nodes associated with a first user and output a first set of abstracted pattern signatures; b) a second data abstraction pipeline configured to receive sensor data from one or more physiological sensing nodes associated with a second user and output a second set of abstracted pattern signatures; and c) a cross-architecture correlation engine configured to compute one or more alignment scores between the first and second sets of abstracted pattern signatures; wherein the alignment scores represent multi-dimensional physiological compatibility and wherein the correlation engine operates exclusively on abstracted representations such that raw sensor data from either user cannot be reconstructed.
2. The system of statement 1, wherein the first user and the second user have differing physiological architectures and use different combinations of sensing nodes.
3. The system of statement 1, wherein the cross-architecture correlation engine performs dimensional mapping by projecting abstracted metrics from each user into a common normalized feature space.
4. The system of statement 1, wherein the alignment scores comprise a composite alignment index, per-dimension sub-scores, and confidence indicators.
5. The system of statement 1, wherein cross-user correlation is gated by mutual consent from both users and supports progressive disclosure of alignment detail.
6. The system of statement 1, further comprising a longitudinal correlation module configured to track alignment scores over time and generate alignment trend indicators.
7. A method comprising: a) receiving a first set of abstracted physiological pattern signatures from a first multi-node sensing ecosystem; b) receiving a second set of abstracted physiological pattern signatures from a second multi-node sensing ecosystem; c) performing cross-architecture correlation comprising dimensional mapping, temporal phase alignment, and pattern similarity scoring; and d) outputting one or more privacy-preserving alignment scores representing multi-dimensional compatibility between the first and second users.
8. The method of statement 7, wherein the cross-architecture correlation is conditioned on systemic context data from an upper-body context node associated with at least one of the users.
9. The method of statement 7, further comprising generating a composite symbolic representation of the alignment between the two users, wherein the symbolic representation contains no individually identifiable physiological measurements.
10. The method of statement 7, wherein the alignment scores are computed locally on a user device using encrypted profile exchange without centralized data aggregation.